TED(10) – 5055

(REVISION-2010) **MODEL QUESTION PAPER**

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

(Branch BME only)

LASER INSTRUMENTATION

(**Maximum Marks: 100**) TIME: **3HRS**

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks.**

I.

1. Define Coherence?
2. List the Q- switching devices?
3. Describe photochemical interaction?
4. Give applications of lasers?
5. Define mode locking? 5**x2=10 marks**

PART - B

Answer any ***FIVE*** of the following questions. Each question carries **6 marks**.

II.

1. Write a short note on lasing mechanism of Ruby laser?
2. Differentiate between temporal and spatial coherence?
3. Define the terms intensity and polarization?
4. List the optical properties of tissue?
5. Define holography and give its applications?
6. Explain construction of Argon laser?
7. With the help of examples summarize application of CO_2 laser?

5x6=30 marks

PART - C

Answer one full question from each Unit. Each question carries 15 marks.

UNIT I

III a) State and explain the properties of lasers? (8)

b) Explain the principle of four level laser? (7)

OR

IV. a) Describe the principle of laser emission with the help of an example? (15)

UNIT II

V. a) Explain the mechanism of Q-switching and shutters used in Q-switching? (15)

OR

VI a) List and explain the mechanisms of mode locking? (15)

UNIT III

VII a).Explain the thermal effects of laser beam? (10)

b) Explain the measurement of skin characteristics? (5)

OR

VIII a) Explain mechanical effects of laser on tissue? (10)

b) Explain laser induced changes in tissue characteristics? (5)

UNIT IV

IX . a) With the help of a schematic diagram explain the construction of Nd- YAG laser? (15)

OR

X a) Explain the construction of ruby laser? (8)

b) Draw and explain the construction of He-Ne laser? (7)

(4 X15=60)